

Vivek Vikram Singh

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PROFESSIONAL SUMMARY

Data Analyst with 3 years at Accenture specializing in SQL, Python, and Power BI, delivering system redesigns, dashboards, and process improvements that drove operational efficiency for a Fortune 500 P&C insurance client.

EDUCATION

University of Massachusetts, Amherst (Isenberg School of Management)

September 2024 - May 2026

Master of Science in Business Analytics

GPA: 3.9/4.0

Manipal University Jaipur, Rajasthan, India

July 2017 - July 2021

Bachelor of Technology in Electronics and Communication Engineering

EXPERIENCE

Business Architecture Analyst (Data & Analytics), Accenture, India

June 2021 - July 2024

- Built a centralized form inventory system on AWS RDS (MySQL) that cut quarterly processing time by 40% (4 Weeks) across 3+ business units during a Waterfall-to-Agile transition, by designing a 12-table schema and PowerApps requirements to consolidate thousands of production forms.
- Developed complex SQL queries (joins, aggregations, window functions) on AWS-hosted datasets to analyze operational records, monitor KPIs and SLA compliance, contributing to \$300K in cumulative operational efficiency gains.
- Built automated Python scripts to extract and clean operational data from SQL databases and shared drives, replacing manual Excel-based cleaning and saving 4-5 hours per cycle for consistent dashboard reporting.
- Performed EDA on 50K+ claims records in Python, replacing Excel-based analysis, identifying 15% data quality gaps and implementing automated validation checks to improve accuracy for leadership decisions.
- Designed and deployed Power BI dashboards for 25+ Accenture and client users across 5 departments, migrating from manual Excel reporting and pulling from AWS RDS to provide leadership visibility into workflows and turnaround times.
- Partnered with business, compliance, and operational stakeholders to gather requirements, draft BRDs, and define reporting logic using structured working sessions and user flow diagrams to reduce ambiguity and rework.

PROJECTS

Airbnb Review Sentiment & Pricing Analytics

February 2026 - May 2026

- Analyzed 60K Airbnb reviews and listing records using NLP techniques including TF-IDF, LDA topic modeling, and DistilBERT sentiment scoring to extract guest experience signals from unstructured text
- Engineered 100+ features from categorical, geospatial, and text-derived data, built and compared Ridge Regression and Neural Network (Keras) models to predict listing prices, achieving R^2 of 0.58
- Derived actionable pricing insights by correlating average sentiment scores with listing prices across Vienna neighbourhoods, identifying key drivers of revenue performance

Customer Analytics & KPI Dashboard

February 2026 - May 2026

- Segmented \$500K+ e-commerce revenue across 8 categories and 4 channels, performing Recency-Frequency-Monetary (RFM) segmentation on 734 customers to identify 119 Champions (31.5% revenue) and 48 At-Risk high-value customers.
- Assessed marketing ROI across 6 advertising platforms, calculating Return on Ad Spend (ROAS) by platform and channel to recommend budget reallocation strategies for improved customer acquisition efficiency and profitability optimization.

Consumer Lending Risk Analysis

September 2025 - December 2025

- Examined 40K Lending Club loan records using Python to identify default risk patterns across borrower profiles and credit grades, detecting high-risk segments with default rates 10-16% above average.
- Created data visualizations using Matplotlib and Seaborn to communicate risk patterns across DTI ratios, employment characteristics, and loan purposes, enabling targeted mitigation strategies.

ACHIEVEMENTS

Pinnacle Award - Q1'24: Recognized for auditing and documenting 3,000+ production forms across Auto, Property, PAF, and PLUS product lines to establish a centralized inventory for the Central Support Unit.

Trailblazer Award - Q2'23: Awarded for building a centralized form inventory system on AWS RDS that cut quarterly processing time by 40% across 3+ business units during a Waterfall-to-Agile transition.

TECHNICAL SKILLS

Technical Stack: SQL (PostgreSQL, SQL Server, MySQL - joins, CTEs, window functions, subqueries), Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), Power BI (DAX, Power Query), Tableau, Excel (Advanced), Power Apps.

Analytics & Methods: Data Visualization, EDA, A/B Testing, Cohort Analysis, RFM Segmentation, Statistical Analysis, Predictive Modeling (Regression, Forecasting), Data Validation, KPI Development, Trend Analysis, Root Cause Analysis.

Cloud & Business Tools: AWS RDS, ETL, Workflow Automation, Stakeholder Management, Agile/Jira, Requirements Gathering, Process Improvement, Data Modeling (relational schema design, star schema).